

B.Tech II Year II Semester (R20) Supplementary Examinations November 2024

PROBABILITY THEORY & STOCHASTIC PROCESSES

(Electronics & Communication Engineering)

Time: 3 hours

Max. Marks: 70

PART – A

(Compulsory Question)

1 Answer the following: (10 X 02 = 20 Marks)

- (a) State the theorem on total probability. 2M
- (b) Define random variable? Classify the random variable with example. 2M
- (c) What is the significance of the probability density function (PDF) for a continuous random variable? 2M
- (d) Define statistical independence between two random variables. 2M
- (e) List the steps involved to calculate the expected value of a function of random variables. 2M
- (f) Define the joint characteristic function for multiple random variables. 2M
- (g) Distinguish between stationary and non-stationary random processes. 2M
- (h) Define ergodicity of random process. 2M
- (i) Discuss the significance of convolution in understanding how linear systems process random signals. 2M
- (j) Define band pass, band-limited, processes in the context of system responses. 2M

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

- 2 (a) State and prove the theorem of total probability. 5M
- (b) In a game of dice, a shooter can win outright if the sum of the two numbers showing up is either 7 or 11, when two dice are thrown. What is the probability of winning outright? 5M

OR

- 3 Of three men, the chances that a politician, a business man or an academician will be appointed as a vice-chancellor of a university are 0.5, 0.3 and 0.2 respectively. Probability that research is promoted by these persons, if they are appointed as vice-chancellor is 0.3, 0.7 and 0.8 respectively. 10M
- (i) Determine the probability that research is promoted.
- (ii) If research is promoted, what is the probability that vice chancellor is an academician.

- 4 (a) Define CDF. State and prove the properties of CDF. 5M
- (b) The Pdf of X is, $f_X(x) = \begin{cases} Kx^2(1-x^3); & 0 < x < 1 \\ 0, & \text{Otherwise} \end{cases}$ 5M

is a valid density function. Find the constant 'k' and cumulative distribution function of x.

OR

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- 5 (a) Define moment generating function. State and prove the properties of MGF. 5M
 (b) Define moment generating function. State and prove the properties of MGF. The exponential density function is given by 5M

$$f_x(x) = \frac{1}{b} e^{-\frac{(x-a)}{b}} ; x > a$$

$$= 0 ; x \leq a$$

Find out variance.

- 6 (a) Define joint distribution function and State & prove the properties of joint distribution function. 5M
 (b) The joint Pdf of random variables X and Y is, 5M

$$f_{XY}(x, y) = \begin{cases} 8xy & ; 0 \leq x \leq 1, 0 \leq y \leq x \\ 0 & ; \text{otherwise} \end{cases}$$

Find $f\left(\frac{y}{x}\right)$ and $f\left(\frac{x}{y}\right)$.

OR

- 7 (a) Explain jointly Gaussian density function for two random variables X and Y. 5M
 (b) Write short notes marginal distributions and marginal density functions of random variables X and Y. 5M

- 8 (a) Define stationary random process and classify the stationary random process. 5M
 (b) Let $N(t)$, be a zero mean, wide sense stationary noise process for which $R_{NN}(\tau) = \frac{N_o}{2} \delta(\tau)$, 5M
 where $N_o > 0$ is a finite constant. Determine whether $N(t)$ is a mean ergodic or not?

OR

- 9 (a) Determine mean and mean square value of system response. 5M
 (b) $X(t) = 2A \cos(\omega_o t + 2\theta)$ is a random process, where ' θ ' is a uniform random variable, over $(0, 2\pi)$. Check the process for stationary. 5M

- 10 Explain the following: 10M
 (i) Effective noise temperature.
 (ii) Average noise figure.

OR

- 11 Derive the expression for effective input noise temperature of cascaded system in terms of their individual input noise temperatures. 10M

B.Tech II Year II Semester (R20) Regular & Supplementary Examinations April/May 2024
PROBABILITY THEORY & STOCHASTIC PROCESSES
 (Electronics & Communication Engineering)

Time: 3 hours

Max. Marks: 70

PART – A
 (Compulsory Question)

1 Answer the following: (10 X 02 = 20 Marks)

- | | |
|---|----|
| (a) State the Baye's theorem of probability. | 2M |
| (b) Define conditional density function and state the properties of it. | 2M |
| (c) Differentiate between central moments and moments about the origin. | 2M |
| (d) State the Central Limit Theorem. | 2M |
| (e) Define moments about the origin for a random variable. | 2M |
| (f) Write short notes on statistical independence for two random variables. | 2M |
| (g) Define stationarity in the context of random processes. | 2M |
| (h) Define Power Spectral Density of random process. | 2M |
| (i) Write short notes on noise figure. | 2M |
| (j) What is thermal noise? How is it quantified? | 2M |

PART – B
 (Answer all the questions: 05 X 10 = 50 Marks)

- | | |
|--|----|
| 2 (a) Discuss Joint and Conditional Probability. | 5M |
| (b) When two dice are thrown, determine the probabilities from axiom 3 for the following events. | 5M |
| (i) $A = \{\text{Sum} = 7\}$, | |
| (ii) $B = \{8 \leq \text{Sum} \leq 17\}$, | |
| (iii) $C = \{10 < \text{Sum}\}$, | |
| (iv) $P(B \cap C)$, | |
| (v) $P(B \cup C)$. | |

OR

- | | |
|--|----|
| 3 (a) State and prove the Baye's theorem of probability. | 5M |
| (b) A speaks truth in 75% and B in 80% of the cases. In what percentage of cases are they likely to contradict each other narrating the same incident? | 5M |
| 4 (a) Define Characteristic function? State and prove the properties of characteristic function. | 5M |
| (b) The characteristic function for a random variable X is given by, $\Phi_X(\omega) = \frac{1}{(1 - j2\omega)^{\frac{N}{2}}}$. Find the mean and second moment of X. | 5M |

OR

- | | |
|---|-----|
| 5 If the random variable X has the MGF, $M_X(t) = \frac{2}{2-t}$. Determine the variance of X. | 10M |
| 6 (a) Define joint Pdf and State & prove the properties of joint probability density function. | 5M |
| (b) Statistically independent random variables X and Y have moments $m_{10} = 2, m_{20} = 14, m_{02} = 12$ and $m_{11} = -6$. Find the moment μ_{22} . | 5M |

OR

Contd. in Page 2

- 7 Consider random variables Y_1 and Y_2 related to arbitrary random variables X and Y by the coordinate rotation, $Y_1 = X\cos\theta + Y\sin\theta$; $Y_2 = -X\sin\theta + Y\cos\theta$. 10M
(i) Find the covariance of Y_1 and Y_2 ,
(ii) For what values of θ , the random variables Y_1 and Y_2 are uncorrelated.
- 8 Define auto correlation function of random process? State and prove the properties of ACF of random process. 10M
- OR**
- 9 State and prove the properties of Power spectral density of random process. 10M
- 10 Explain the following: 10M
(i) Extraterrestrial noise,
(ii) Shot noise,
(iii) Thermal noise.
- OR**
- 11 Derive the expression for effective input noise temperature of cascaded system in terms of their individual input noise temperatures. 10M

B.TechII Year II Semester (R20) Regular &Supplementary Examinations August/September2023

PROBABILITY THEORY & STOCHASTIC PROCESSES

(Electronics & Communication Engineering)

Time: 3 hours

Max. Marks: 70

PART – A

(Compulsory Question)

1 Answer the following: (10 X 02 = 20 Marks)

- | | |
|--|----|
| (a) Define probability using the axiomatic approach. | 2M |
| (b) What are the conditions for a function to be a random variable? | 2M |
| (c) What are equal and unequal distributions? | 2M |
| (d) Define moment generating function and write the formula to find mean and variance. | 2M |
| (e) List the properties of Gaussian random variables. | 2M |
| (f) Define Joint central movement. | 2M |
| (g) What is random process? | 2M |
| (h) Explain First order stationary process briefly. | 2M |
| (i) Define autocorrelation function of response. | 2M |
| (j) Define colored noise. | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

- 2 (a) An experiment consists of observing the sum of the outcomes when two fair dice are thrown. Find the probability that the sum is 7 and find the probability that the sum is greater than 10. 5M
- (b) In a factory there are 4 machines produce 10%,20%,30%,40% of items respectively. The defective items produced by each machine are 5%,4%,3% and 2% respectively. Now an item is selected which is to be defective, what is the probability it being from the 2nd machine? 5M

OR

- 3 (a) What is conditional distribution? What are the methods of defining conditioning event? 5M
- (b) Describe different types of Random variables with examples. 5M
- 4 (a) Derive the Moment generating function About Origin. 5M
- (b) Calculate mean, standard deviation, moment in each of the following alternative cases: 5M
- (i) Number of trials are 18 and probability of success is $1/3$,
- (ii) Number of trials are 18 and probability of failure is $1/3$.

OR

- 5 (a) State and explain the properties of joint density function. 5M
- (b) The joint density function of random variables X and Y is; 5M
- $(X, y) = \{ 8xy; 0 \leq x < 1, 0 < y < 1$
- Otherwise $x, y = 0$.
- Find $f(y/x)$ and $f(x/y)$.

Contd. In Page 2

- 6 (a) Write short notes on Gaussian distribution and also find its mean. 5M
(b) Consider that a fair coin is tossed 3 times, Let X be a random variable, defined as X = number of tails appeared, find the expected value of X . 5M
- OR**
- 7 (a) Explain about Linear transformations of Gaussian random variables. 5M
(b) Explain point conditioning and interval conditioning. 5M
- 8 A random process has sample functions of the form $X(t)=A\cos(\omega t+\theta)$ where ω is constant, A is a random variable with mean zero and variance one and θ is a random variable that is uniformly distributed between 0 and 2π . Assume that the random variables A and θ are independent. Is $X(t)$ mean ergodic-process? 10M
- OR**
- 9 Explain about the following random process: 10M
(i) Mean ergodic process,
(ii) Correlation ergodic process,
(iii) Gaussian random process.
- 10 Derive the relationship between crosspower spectral density and cross correlation function. 10M
- OR**
- 11 For a input output linear system($X(t), h(t), Y(t)$), derive the cross correlation function $R_{xy}(\tau)$ and the output auto correlation function $R_{xx}(\tau)$. 10M
